GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

k = 3

copy ÷ original

Scoring: 1 point.

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.5

copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☒ **B. Each side is 2 times as long.**
- ☒ **C. The perimeter is 2 times as long.**
- ☒ **D. The area is 4 times as large.**
- ☐ E. Each angle is 2 times as large.

B, C and D

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

32 cm

perimeter × k

b. Find the copy's area. *1 pt*

60 cm²

area × k² — 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 6 cm². Its scale copy has area 150 cm². What is the scale factor? *1 pt*

k = 5

150 ÷ 6 = 25 is k², so k = √25. A board with 25 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

10 ft

the scale per 1 inch, then ×: 1 in : 5 ft

b. Find the room's real area. *1 pt*

150 ft²

real dimensions first: 10 by 15 — 6 in² × 5 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

135 ft²

9 ft by 15 ft.

Scoring: 1 point.

Converting Across Systems

8 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

17.78 cm (17.9 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☒ **B. 110 × 0.39**
- ☐ C. 110 ÷ 0.39
- ☒ **D. 110 ÷ 2.54**

B and D

Both sheet routes (42.9 and 43.3) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

137.16 cm

a chain: 54 in, then × 2.54 (54 ÷ 0.39 = 138.5 by the other ratio — full credit). The estimate catches 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

77.14 lb (77.09 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

11.355 L (11.36 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 in = 2.54 cm. A tile is 10 in^2 in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 cm^2
- ● **B. 64.5 cm^2**
- ○ C. 6.45 cm^2
- ○ D. 50.8 cm^2

B

$10 \times 2.54^2 = 64.52$ (or $10 \div 0.39^2 = 65.7$ — the sheet's two routes differ here; either picks B).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

a. How many Canadian dollars is $90? *1 pt*

118.80 Canadian dollars

× the rate

b. How many US dollars is 66 Canadian dollars? *1 pt*

$50

÷ the rate; 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 27 miles per gallon and its tank holds 10 gal.

a. How far does it go on a full tank? *1 pt*

270 mi

the rate × the gallons

b. It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

5 tanks (1,200 ÷ 270 ≈ 4.44)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 5 servings uses 2 cups of rice.

a. How much rice for 15 servings? *1 pt*

6 cups

k = 3 from the yields, then the ingredient

b. With 10 cups on hand, how many servings can be made? *1 pt*

25 servings

5 recipes × the yield; 50 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

24 gal

k = 4 from the second color, then the whole, 6 parts; 12 found the first color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.

Same unit first; 80 > 45 compared two units.

Scoring: 1 point.